Problem 18.22

A perpetuity pays 1 at the end of every year plus an additional 1 at the end of every second year. The present value of the perpetuity is K for $i \geq 0$. Determine K.

Solution

The perpetuity can be split into two streams of payments:

1. 1 at the end of every year, and
2. an additional 1 at the end of every second year.

The present value of the first stream is

$$a_{\overline{\infty}|i} = \frac{1}{i}.$$

For the second stream, payments occur every 2 years. Let j be the effective interest rate for a 2-year period. Then

$$1 + j = (1+i)^2,$$

so

$$j = (1+i)^2 - 1 = 2i + i^2 = i(2+i).$$

The present value of this second perpetuity is therefore

$$a_{\overline{\infty}|j} = \frac{1}{j} = \frac{1}{i(2+i)}.$$

Hence,

$$K = a_{\overline{\infty}|i} + a_{\overline{\infty}|j} = \frac{1}{i} + \frac{1}{i(2+i)}.$$

Combining terms,

$$K = \frac{2+i}{i(2+i)} + \frac{1}{i(2+i)} = \frac{i+3}{i(2+i)}.$$

Thus,

$$\boxed{K = \frac{i+3}{i(i+2)}}.$$